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The price elasticity of marijuana demand: evidence from crowd-sourced transaction data

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Abstract This paper uses crowd-sourced transaction data from a cross section of the USA to examine demand for marijuana. State and regional variations in consumption, price, and quality are also explored. Our data are a unique cross section of over 23,000 actual marijuana transactions where price, quantity, and quality are reported, allowing for an estimation of the full demand elasticity rather than the participation elasticity. In addition, we account for the endogeneity of price by using instrumental variable estimation to calculate price elasticity. Price elasticity of demand estimates ranges between −0.67 and −0.79. Noticeable price differences are found between high-, medium-, and low-quality marijuana, with high-quality marijuana, at $13.77 per gram, 144 % greater than low-quality marijuana, at $5.63 a gram. Significant price variation is also found by medical marijuana status and census region, although this variation depends critically on the quality of the marijuana.

Keywords Elasticity · Marijuana demand · Instrumental variable estimation

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1 Introduction

Over forty years have passed since Nisbet and Vakil (1972) surveyed UCLA students in an attempt to estimate the own-price elasticity of demand for marijuana. Their

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precedential study paved the way for a topic that has grown exponentially in importance as states recently have decriminalized possession, passed medical marijuana laws, and, in Colorado, Washington, Alaska, and the District of Columbia, passed legislation legalizing possession for personal use.[1] The motivation for such policies, particularly those in Colorado and Washington, has been to legalize possession in order to use police resources against more serious property and violent crimes, regulate the industry and remove illegal drug dealers from its provision, tax sales in order to raise revenue, and treat addiction as a public health issue instead of a criminal issue. The impact of these laws, including the response from the Federal Government which still considers marijuana possession illegal, remains to be seen. The changes in demand and supply that result as well as the tax revenue generation potential of marijuana sales have also yet to be determined. These impacts will depend at least in part on the price elasticity of demand. The drastic changes that have occurred with marijuana legislation since Nisbet and Vakil (1972) estimated the price elasticity, and the fact that very few studies have estimated the price elasticity since, is the motivation for this study.

According to the Substance Abuse and Mental Health Services Administration of the Department of Health and Human Services (2012), marijuana is the most commonly used illicit drug with 18.1 million past month users. Between 2007 and 2011, the rate of use has increased an estimated 5.8 to 7.0 % (Substance Abuse and Mental Health Services Administration 2012). Marijuana use has been a hot policy topic in the USA for many years. The debate generally centers on the contention that legalization of marijuana would result in sky rocketing consumption versus the futility of the "war on drugs" with claims that legalization and taxation is the rational solution.

This paper does not seek to enter that debate. Rather, the purpose of this paper is to provide an estimate of the own-price elasticity of demand using data on price, quantity, and quality from a wide cross section of marijuana transactions. Clearly, this will be of interest to many as changes in the price of marijuana, perhaps due to changes in its legal status, will impact its consumption.

The next section of this paper reviews the extant literature on marijuana demand. Surprisingly, few studies estimate the elasticity of demand, primarily due to the lack of available data. The third section describes the crowd-sourced data used in this study. This is followed by our empirical methodology and results. The last section concludes the paper.

2 Literature review

Clements (2009) demonstrates that the conventional price elasticity of demand, $\partial \log q / \partial \log p$, is the sum of the elasticity of the number of users, also known as the participation elasticity, and the elasticity of demand of existing users. Estimating the elasticity of demand for an illicit good such as marijuana, be it the conventional elasticity or the participation elasticity, is problematic since rarely is information about

[1] On November 6, 2012, Colorado voters passed Amendment 64 and Washington voters approved Initiative 502; Alaska voters approved Ballot Measure 2 and the District of Columbia passed Initiative 71 on November 4, 2014.

purchases systematically recorded. Consequently, data are frequently gathered from surveys asking participants about their prevalence and frequency of marijuana use in the past month or year. The probability of consuming is then estimated based on various socioeconomic (e.g., gender and race) or policy variables (e.g., decriminalization) and, less frequently, price. Studies of this type, several of which are reviewed below, estimate how price impacts the probability of using marijuana, thereby providing an estimate of the *participation* elasticity, not the conventional elasticity. Our study is one of the few that has data on price and quantities purchased, providing an estimate of the conventional price elasticity of demand.

The first publication investigating the marijuana price elasticity was conducted by Nisbet and Vakil (1972). Their data were gathered from an anonymous mail questionnaire of 926 UCLA students. Students reported quantities of marijuana purchased at current prices and also quantities they would purchase in the future after hypothetical price changes. Nisbet and Vakil (1972) employed ordinary least squares (OLS) to estimate the effect of price and monthly expenditures on the quantity demanded. Both the double log and linear functional forms were estimated. The researchers concluded that the conventional price elasticity of marijuana is between −0.40 and −1.51 depending on the data and the functional form.

Due to the lack of reliable transactions data, particularly price data, many subsequent marijuana demand studies focused on other determinants of prevalence or frequency of use. While a thorough review of these studies is beyond the scope of the current analysis (see, however, Pacula and Lundberg 2014), these determinants include decriminalization (e.g., Model 1993; Saffer and Chaloupka 1999; Williams 2004) and price changes in related goods, including alcohol (e.g., Thies and Register 1993; Pacula 1998; Cameron and Williams 2001; DiNardo and Lemieux 2001), tobacco (Zhao and Harris 2004), and other illicit drugs such as cocaine (e.g., Saffer and Chaloupka 1999; Desimone and Farrelly 2003).[2] Some of these studies include marijuana price data, yet most do not provide marijuana price elasticity estimates in part due to poor availability of price data.

One source for marijuana price data is the System to Retrieve Information from Drug Evidence (STRIDE), maintained by the Drug Enforcement Administration (DEA) of the US Department of Justice. As noted by Pacula et al. (2001), however, these data do not contain information on quality or distinguish between wholesale and retail prices and are therefore of limited use for marijuana studies. Moreover, Caulkins and Pacula (2006) note that STRIDE data represent marijuana transactions between strangers which represent a small fraction of total transactions since most transactions occur between friends. Another source of marijuana price data is the Illegal Drug Price/Purity Report (IDPPR), also maintained by the DEA. While containing information on drug purity and distinguishing between wholesale and retail prices, this data source only contains price for nineteen cities in sixteen states.

Since the publication of Nesbit and Vakil (1972), only a handful of other studies have estimated any marijuana elasticity, with most examining how price impacts 30-day or annual use rates of youth. One of the earliest post-Nesbit and Vakil studies conducted in

[2] See Pacula and Lundberg (2014) for an in depth exportation of the existing literature.

the USA is Pacula et al. (2001). Using both national and repeated cross-sectional data sets from Monitor the Future (MTF) and marijuana price data from the IDPPR, Pacula et al. (2001) estimate the participation elasticity of marijuana demand for youth. Their estimates range from a low of −0.002 to −0.69. The range of estimates depends on the inclusion of potency and time trends as independent variables. Given concerns over the potential endogeneity of other independent variables, including the perception of harm, the preferred specification of Pacula et al. (2001) yields a participation elasticity of annual and 30-day use of −0.30, suggesting a 10 % reduction in price would increase the number of youth reporting past 30-day and annual use by 3 %.

In Australia, van Ours and Williams (2007) use self-reported data on youth marijuana use to estimate how price and other factors influence the decision to begin and stop using marijuana. Price data for their study are from the Australian Illicit Drug Reports (AIDR), which are gathered through drug enforcement agents similarly to IDPPR. Price data are not available for every Australian state in every time period, however, so van Ours and Williams (2007) impute prices using the method employed in Saffer and Chaloupka (1999). van Ours and Williams estimate initiation elasticities ranging from −0.31 to −0.70, with their preferred model including state fixed effects and a time trend yielding an elasticity of −0.50, suggesting a 10 % reduction in price increases the number of youth starting to use marijuana by 5 %. Price was not found to significantly influence the decision to quit using marijuana.

Other studies have estimated the participation elasticity for regular marijuana users, defined as those who report use on a monthly basis (Pacula and Lundberg 2014). For example, Williams et al. (2004) use IDPPR price data and estimate a participation price elasticity of −0.24 for college students in the USA. A summary of additional studies can be found in Pacula and Lundberg (2014) as well as Gallet (2014) who conducts a meta-analysis of numerous illicit drug studies and finds that the elasticity for marijuana ranges between −0.15 and −0.31, which is less than that of heroin and cocaine. Gallet (2014) finds that studies using experimental data estimate significantly higher elasticity estimates, which may explain the high elasticity estimates found by Nisbet and Vakil (1972). In addition, Gallet (2014) finds that the participation elasticity exceeds the elasticity for existing users, that elasticity estimates are lower in the USA, and that accounting for the price of goods that serve as complements or substitutes to the drug in question significantly influences elasticity estimates. However, Gallet (2014) does not account for potency of the drug in his meta-analysis, a factor that might clearly influence the quantity, price, and demand elasticity of a drug consumed.

As noted above, the conventional, full demand elasticity is the sum of the participation elasticity and the elasticity for existing users. The studies reviewed above estimate the participation elasticity but provide little if any information on the price responsiveness of existing users. Yet, the vast majority of marijuana consumed is by existing, heavy users, particularly daily users. For example, Caulkins and Pacula (2006) show that 26 % of users account for 71 % of total purchases. Similarly, Kilmer et al. 2014 find that daily users account for the vast majority of expenditures even though they are not the majority of users. Consequently, studies estimating only the participation elasticity are missing the price responsiveness of the vast majority of marijuana purchases.

Since the publication of Nisbet and Vakil (1972), very few studies have estimated the full demand elasticity. Clements (2009), summarizing and building upon work

by Clements and Daryal (2005) and Clements (2004, 2008), find a conventional, full demand price elasticity of −0.4 in Australia. Elasticity does, however, vary by type of user and gender. Clements (2009) notes that the estimate of −0.4 is based on a 50 % price reduction and that a smaller, more marginal reduction in price would result in a larger elasticity. Clements (2008, 2009) therefore posits that a useful rule of thumb is that the price elasticity is approximately −0.5. Given the potential measurement error in marijuana price and use data, Clements et al. (2010) use a Monte Carlo simulation procedure to account for this uncertainty and, among other findings, arrive at price elasticity estimates for marijuana ranging from −0.586 to −0.66.

The current study builds on and contributes to the existing literature reviewed above in several key ways. First, to our knowledge, this is the only study to empirically estimate the conventional, full demand price elasticity of marijuana demand using actual transaction histories reported directly from a large number of consumers. While similar to Nisbet and Vakil (1972) in that the data are self-reported, our sample size is notably larger with over 23,000 transactions. Second, it also contains transaction data from every state in the USA, making the imputation of missing data unnecessary while providing a clearer picture of geographic variation in price. Moreover, the data reflect private transactions from individual consumers rather than prices paid by drug enforcement officers, as in the STRIDE, IDPPR, and AIDR data sets. Although it is generally agreed that undercover agents pay a realistic price so as not to reveal their identity to the illegal drug dealer, the data used in this study reflect a broader cross section of purchasers from all over the USA. Third, the current data set also allows us to control for variations in quality. Fourth, we confirm the robustness of our results using instrumental variable estimation to account for the endogeneity of price in the demand equation.

3 Data and empirical methodology

The data set used in this study is a series of transaction data obtained from the creator of the Web site called "The PriceOfWeed.com." This Web site was designed to answer one simple question: "What is marijuana really worth?" The creators set out to "crowd-source" the street value of marijuana from the most accurate source possible, the consumer. Site visitors anonymously input the following about their most recent marijuana transactions: amount purchased; price of the purchase; and quality, choosing from low, medium, or high.[3] Data on the city, state, and country where the transaction took place are also collected. Data from over 31,000 transactions were obtained between September 2, 2010, and August 29, 2011.[4] After dropping observations for transactions that took place outside of the USA (6684 observations), observations with prices exceeding $100 per gram (47 observations), observations with prices of zero (47 observations), and transactions from Alaska and Hawaii (154 observations), we are

[3] The Web site has a drop-down menu for quantity with the following options: 1 gram, 5 grams, 10 grams, 1/8 ounce, 1/4 ounce, 1/2 ounce, or 1 ounce. The quality field is also a drop-down menu with low, medium, and high options. Prices are entered freely.

[4] Malivert and Hall (2013) also use data from www.priceofweed.com, though they only use one month of data and do not report price elasticity.

Table 1 Descriptive statistics

Variable	Mean	SD	Minimum	Maximum
Price ($ per gram)	11.518	5.972	0.101	100
Quantity (grams)	13.920	11.066	3.5	28
Quality high	0.600	0.489	0	1
Quality medium	0.332	0.471	0	1
Quality low	0.0678	0.251	0	1
Medical marijuana state	0.333	0.471	0	1
Decriminalized state	0.364	0.481	0	1
Income (dollars)	40365.19	5542.03	30426	68013
Beer tax ($ per gallon)	0.247	0.225	0.019	1.05
Electricity price ($ per kilowatt hour)	12.569	3.035	7.95	19.33

Excludes Alaska and Hawaii and observations where price is zero

left with 23,611 usable observations.[5] Summary statistics for these data are provided in Table 1.

The quantity of marijuana is measured in grams. Marijuana is typically packaged in specific amounts: eighths (3.5 grams), quarters (7 grams), half ounces (14 grams), and ounces (28 grams). Observations include amounts between 3.5 grams and 28 grams. The limited range of reported quantities should minimize any price differences stemming from quantity discounts such as those found by Clements (2004). The average purchase is just <14 grams, with the median purchase amount at 7 grams. Price is dollars per gram, with an average price per gram of $11.52.

Table 2 provides information on the total number of observations, number of observations in each quality category, and average prices and quantities by state, including Alaska and Hawaii. The highest average price, $15.13 per gram, is in Washington, D.C., whereas the lowest, $8.84, is in Oregon. South Dakota and Wyoming are least represented in our sample, with 50 observations each, while California is the most represented with 2737 observations.

While these data are unique and rich in certain aspects, several caveats should be noted at this point. Most obviously, these data are self-reported and not a random sample. The self-reported nature of data also makes the stated quality of marijuana for each purchase somewhat subjective. The anonymity of how the data was reported, while likely ensuring more accurate responses, also poses some challenges. Since no individual level data are available, we do not have any demographic or economic information about the individuals involved in these transactions. Moreover, we do not know whether each transaction represents a unique individual, or whether some individuals report multiple transactions over time. Related to this point, we cannot distinguish whether purchases represent new users who have decided to begin using

[5] Prices over $100 a gram and $0 per gram were thought to be unrealistic. Including these observations does not dramatically change our findings. OLS elasticity estimates average -0.530, while the instrumental variable estimates range from -0.656 to -0.783. Similarly, dropping Alaska and Hawaii made no difference in our elasticity estimates, but their inclusion unnecessarily complicates the instruments chosen in our instrumental variable estimation.

Table 2 State marijuana statistics

Region state	Total observations	Mean price per gram	Mean quantity in grams	Number of transactions reporting		
				High quality	Low quality	Medium quality
New England Region						
Connecticut	342	13.33	12.12	202	22	118
Maine	186	11.03	12.84	91	19	76
Massachusetts	921	13.38	13.19	542	60	319
New Hampshire	164	11.94	14.28	89	16	59
Rhode Island	157	12.53	15.78	106	11	40
Vermont	126	12.56	12.33	78	10	38
Mid-Atlantic Region						
New Jersey	594	13.32	13.93	311	34	249
New York	1932	13.28	13.20	1,163	113	656
Pennsylvania	1064	12.00	13.61	604	69	391
South Atlantic Region						
Delaware	70	11.65	13.40	38	8	24
District of Columbia	121	15.13	12.61	80	9	32
Florida	1414	10.76	14.56	864	92	458
Georgia	593	11.29	13.95	333	35	225
Maryland	426	12.66	14.05	258	34	134
North Carolina	624	11.72	13.43	365	48	211
South Carolina	296	10.29	14.42	148	30	118
Virginia	628	12.61	12.31	357	45	226
West Virginia	98	11.19	14.10	42	5	51
East South Central Region						
Alabama	173	10.46	14.52	83	15	75
Kentucky	256	9.26	15.55	110	22	124
Mississippi	96	10.26	16.01	47	17	32
Tennessee	324	11.65	14.19	194	26	104
East North Central Region						
Illinois	1014	13.36	12.49	600	64	350
Indiana	418	9.85	13.19	199	34	185
Michigan	847	10.69	14.55	524	63	260
Ohio	901	9.40	14.28	422	87	392
Wisconsin	458	13.79	10.96	313	21	124
West South Central Region						
Arkansas	151	9.79	16.98	73	21	57
Louisiana	229	11.60	15.98	118	47	64
Oklahoma	179	10.52	15.81	99	29	51
Texas	1315	11.43	15.39	782	164	369

Table 2 continued

Region state	Total observations	Mean price per gram	Mean quantity in grams	Number of transactions reporting		
				High quality	Low quality	Medium quality
West North Central Region						
Iowa	253	11.97	13.71	137	31	85
Kansas	314	11.23	14.92	211	25	78
Minnesota	449	13.73	12.42	312	25	112
Missouri	478	10.90	12.93	258	62	158
Nebraska	159	10.73	13.11	94	17	48
North Dakota	53	14.61	11.59	27	3	23
South Dakota	50	13.52	12.59	33	4	13
Mountain Region						
Arizona	384	10.34	16.67	229	40	115
Colorado	520	10.05	14.60	389	16	115
Idaho	102	11.15	13.78	64	1	37
Montana	193	10.55	12.57	168	0	25
Nevada	151	10.37	14.94	92	3	56
New Mexico	118	10.61	14.71	79	8	31
Utah	163	11.41	13.78	117	7	39
Wyoming	50	13.58	8.16	25	6	19
Pacific Region						
Alaska	78	12.47	19.01	45	1	32
California	2737	10.58	13.70	1,783	64	890
Hawaii	76	13.16	15.36	55	2	19
Oregon	617	8.84	16.01	457	6	154
Washington	715	10.00	15.40	469	16	230

marijuana, regular users, or heavy users. As noted by Pacula and Lundberg (2014), elasticity estimates are likely to vary by type of user. Therefore, our estimates are more similar to Nisbet and Vakil (1972) and Clements (2009) in that changes in consumption due to price variation reflects a full price elasticity of demand as opposed to the participation elasticities found in other studies.

Finally, given the non-random variation in sample size across states, it is possible that certain states may overly influence our elasticity estimates and that our error terms are not independently and identically distributed. Consequently, we estimate weighted OLS and instrumental variable regressions with cluster robust standard errors.[6]

[6] Observations are weighted by the number of observations in each state, specifically the inverse of the probability that each state is on our sample.

Given the lack of individual level socioeconomic data, the remaining variables used in this analysis are measured at the state level. For income, we use average per capita income from the Bureau of Economic Analysis, measured in dollars, for the state in which the transaction took place.

The beer tax is intended to capture the price of a related good. There has been much debate as to whether alcohol is a substitute or a complementary good with respect to marijuana. It is measured by the state excise tax placed on beer in the state where the transaction took place. All excise taxes are measured in dollars per gallon and were obtained from the Tax Foundation.

Quality is measured by dummy variables that capture if the marijuana in the transaction was reported as high, medium, or low quality. In our sample, 60 % of transactions involved marijuana reported as high- quality, 33 % medium-quality, and 7 % low-quality marijuana. It deserves note that this a subjective judgment, and one that may be further complicated by some individuals reporting before use with others reporting after. Nonetheless, as Caulkins and Pacula (2006, page 196) point out, dilution of marijuana is a physical (as opposed to a chemical) process which "can easily be detected through physical inspection…" Given this fact and the large overall number of observations, any error introduced by subjectivity should be sufficiently random as to not bias the results.

Decriminalization is a dummy variable that identifies whether or not the transaction took place in a state in which penalties associated with marijuana are decriminalized. Thirty-nine percent of transactions took place in a decriminalized state. It should be noted, however, that these eleven states, identified in Pacula et al. (2003), have varying penalties associated with possession and some non-decriminalized states also have reduced penalties associated with possession. In fact, Pacula et al. (2003) note that decriminalized states cannot be uniquely identified. We describe how we account for this in more detail below.

To estimate the price elasticity of demand, the following model is estimated using weighted OLS and instrumental variable estimation with cluster robust standard errors:

$$lnQ_i = \beta_0 + \beta_1 lnP_i + \beta_2 lnI_s + \beta_3 lnB_s + \beta_4 H_i + \beta_5 M_i + \beta_6 D_s + \varepsilon_i \qquad (1)$$

where lnQ_i represents the log of quantity of grams purchased in transaction i, $\ln P_i$ represents the log of price paid for that quantity, measured as dollars per gram, $\ln I_s$ is the log of per capita income in the state where the transaction occurred, and $\ln B_s$ is the log of the beer tax, in dollars per gallon, for the state where the transaction occurred. H_i is a dummy variable equal to one if the marijuana purchase was reported as high quality, and M_i is a dummy variable equal to one if the marijuana purchased in transaction i was reported as medium quality. Low quality is the omitted category. D_s is a dummy variable equal to one if marijuana in state s is decriminalized.

This model is also estimated using instrumental variable techniques to account for the endogeneity of price using electricity prices, distance to Mexico City, and the legal status of medical marijuana for supply-side instruments.[7] The sources of these

[7] Because variation in price and quantity may reflect changes in supply as well as demand, price is endogenous in our demand equation, requiring Instrumental Variable estimation to verify that the demand

variables are described here, with the rationale for these instruments discussed in depth in the next section.

Electricity price is the average per capita electricity price ($/kilowatt hour or kWh) for the state in which the transaction took place. This information was provided by the US Energy and Information Administration through the "Monthly Electric Sales and Revenue Report with State Distributions Report."

Distance to Mexico City captures the dependence of the US market on imports from Latin America. Constructed using ArcGIS, it measures the distance from each state's geographic center to Mexico City in kilometers.

The medicinal marijuana variable is a dummy variable that represents whether or not the state in which the transaction took place has legalized medicinal marijuana prior to or during 2010. This information was obtained from the Office of National Drug Control Policy's Marijuana Resource Center.[8] Approximately thirty-four percent of transactions took place in medical marijuana states.

4 Empirical results

4.1 OLS estimates

Table 3 shows weighted OLS results from estimating the demand model given by Eq. (1). All standard errors are cluster robust, clustered at the state in which the transaction took place. The coefficient of the price variable indicates that the demand for marijuana is inelastic and equal to −0.689. This estimate is very much in line with the most recent estimates of van Ours and Williams (2007) and Clements et al. (2010), and higher (in absolute value) than most of the estimated participation elasticities reviewed above. This is an expected result since we are calculating the conventional price elasticity of demand on quantities purchased which is the sum of the participation elasticity and the price elasticity of existing users.

The other control variables are frequently statistically insignificant at conventional levels. One exception is the coefficient on the dummy variable indicating high-quality marijuana was purchased. This coefficient suggests a greater demand for higher- quality marijuana. As the quality measure is somewhat subjective, these results suggest that purchasers place a greater value on marijuana that is perceived as high quality and that marijuana of perceived lower and medium quality may be more substitutable.

The coefficients estimated for the beer tax and decriminalization, although statistically insignificant, are difficult to interpret as the impact of variation in the price of a related goods and state criminalization policy, respectively, since it is not possible to disentangle those impacts from other state-specific fixed effects. In addition, the decriminalized variable is measured with considerable noise since, as noted by Pacula et al. (2003), decriminalized states are not uniquely identifiable since several non-decriminalized states have reduced the criminal status of marijuana possession

Footnote 7 continued
elasticity is accurately estimated. See Angrist and Kreuger (2001) for an overview of instrumental variable techniques.

8 Available at http://www.whitehouse.gov/ondcp/state-laws-related-to-marijuana.

Table 3 Weighted and cluster robust OLS and IV estimates of marijuana demand *dependent variable: log of grams of marijuana purchased*

Variables	OLS (1)	OLS (2)	OLS (3)	IV (4)	IV[a] (5)	IV (6)	IV[a] (7)
Ln (price)	−0.689***	−0.688***	−0.689***	−0.782***	−0.597**	−0.789***	−0.673**
	(0.0210)	(0.0205)	(0.0220)	(0.167)	(0.274)	(0.165)	(0.265)
Ln (income)	0.138	0.143*		0.192*	0.0847	0.202*	0.134
	(0.0827)	(0.0813)		(0.101)	(0.165)	(0.111)	(0.164)
Ln (beer tax)	0.0357	0.0349		0.0349	0.0365	0.0338	0.0350
	(0.0246)	(0.0245)		(0.0235)	(0.0262)	(0.0233)	(0.0251)
Quality high	0.286***	0.284***	0.291***	0.383**	0.189	0.389**	0.268
	(0.0472)	(0.0460)	(0.0497)	(0.182)	(0.288)	(0.178)	(0.280)
Quality medium	−0.0365	−0.0371	−0.0333	0.0100	−0.0827	0.0135	−0.0445
	(0.0371)	(0.0363)	(0.0391)	(0.0904)	(0.139)	(0.0884)	(0.136)
Decriminalized	−0.0350			−0.0419	−0.0281		
	(0.0297)			(0.0301)	(0.0329)		
Constant	2.252**	2.194**	3.654***	1.822**	2.679*	1.712*	2.264
	(0.858)	(0.844)	(0.0338)	(0.917)	(1.404)	(1.006)	(1.404)
State dummies	No	No	Yes	No	No	No	No
R^2	0.217	0.217	0.227				
N	23,611	23,611	23,611	23,611	23,611	23,611	23,611

Cluster robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

[a] IV estimates with only the natural log of electricity price and distance to Mexico City as instruments

offenses either through changing the statutory offense or through conditional discharge and expungement of records for first-time offenders.

Despite the inability to interpret the beer tax and decriminalization variables directly, and their statistical insignificance, we feel it is important to include them in our analysis since they may be capturing unmeasured state variation. As a robustness check, we re-estimate Eq. (1) excluding the decriminalized variable. These results are presented in column 2 and are very similar to those reported when including *Decriminalized*. As a further robustness check, we also used a definition of decriminalized from the National Organization for the Reform of Marijuana Laws (NORML). NORML defines decriminalization as "no jail time or criminal record for first-time possession of a small amount for personal consumption" and at the time of our sample included the eleven original decriminalized states plus Massachusetts and Nevada. These results, not separately reported, were also nearly identical to those reported in columns (1) and (2). As a final robustness check, we estimate the model including state fixed effects, which should capture any differences in punishment and enforcement of marijuana laws which have been shown to be an important consideration when estimating the demand for illicit drugs (Desimone and Farrelly 2003). Income and the beer tax variables are dropped since they are state averages for a single year and consequently are collinear with the state fixed effects. Our estimate of the price elasticity when including state fixed effects remains consistent at −0.689.

4.2 Instrumental variable estimates

To account for the endogeneity of price in our demand equation, we use electricity price, distance to Mexico City, and medical marijuana designation as supply-side instruments. The first two of these instruments are chosen to reflect the main sources of marijuana supply in the USA: domestic production and imports from Mexico (Kilmer et al. 2010).

Domestically, electricity is a large cost factor in domestic production of marijuana. Marijuana production typically takes two forms in the USA: indoor or outdoor growth. Because not many climates in the USA are suitable for marijuana growth, as well as for security reasons, most domestic production of marijuana takes place indoors. Without the power of the sun, marijuana plants require high-intensity discharge (HID) lighting to grow. HID lights require large amounts of electricity and typically need to be on for 12–24 h a day in order to simulate the normal cycles of the sun. Electricity is also required for many of the other production processes such as dehumidification, space heating, and generation of CO^2. Caulkins (2010) estimates that lighting accounts for \$75 of the \$200–\$400 cost per pound of indoor production. Mills (2012) estimates that national electricity use attributable to indoor marijuana growth is somewhere around 20 TWh/year. This is equivalent to the electricity consumption of 2 million average US homes, or 1 % of national electricity consumption.

Most of the rest of the marijuana consumed in the USA enters the USA over the US-Mexican border. In fact, an outright majority of marijuana consumed in the USA may come from Mexico, as Kilmer et al. (2010) estimate that anywhere between 40 and 67 percent of marijuana consumed in the USA comes from Mexico. To account for this supply factor, the distance from each state's geographic center to Mexico City is employed as an instrument. This instrument is appealing because of its simplicity, consistency, and exogeneity. The US-Mexico border is long with multiple major ports of entry. One could alternatively use state distance to the nearest major border crossing as an instrument, but this would require assuming that: (1) marijuana being smuggled into the USA comes through the major ports of entry, and (2) the marijuana consumed in a given state comes from the closest border crossing. By using a single point for reference in the interior of the country, no assumptions about the transportation or distribution networks need to be made while acknowledging the simple fact that a large portion of marijuana consumed in the USA enters from Mexico.

Our other instrument to identify the demand function is the policy variable indicating whether or not marijuana has been legalized for medicinal purposes in the transaction state. It is true that this policy variable is probably correlated with a consumer's willingness to purchase marijuana in that it may be associated with reduced risk of use and increased social acceptance and will therefore likely influence demand. However, whereas the decriminalization variable decreases risk for possession of small amounts, medical marijuana legislation is much more relevant for the producer. A medicinal marijuana prescription not only allows patients to possess and consume marijuana but allows for home cultivation (Pacula et al. 2010). In California alone, over 400,000 individuals are legally authorized to grow marijuana for medical use (Mills 2012). Nevertheless, because the medical marijuana policy theoretically may

impact both demand and supply, we estimate our model both with and without this instrument as a robustness check.

Instrumental variable (IV) estimates of Eq. (1) are provided in columns (4) through (7) of Table 3. In addition to estimating the equation with and without medical marijuana as an instrument, columns (6) and (7) report the estimates excluding *Decriminalized* for consistency with the OLS estimates. All estimates are from weighted IV with cluster robust standard errors. First-stage estimates are provided in Table 5 in the Appendix. The point estimates for the elasticity of price are either similar or slightly larger to those obtained when using OLS, ranging from −0.597 to −0.789. These estimates are slightly larger in magnitude (in absolute value) than Clements (2008, 2009) findings that price elasticity for marijuana is −0.5, but similar to Clements et al. (2010). Not surprisingly, given that we are using IV and robust standard errors, the standard errors are noticeably larger than the OLS counterparts, giving a 95 % confidence interval for our elasticity estimates that range from approximately −0.06 to −1.19. With the larger standard errors, several of the additional control variables are not statistically significant, although a greater demand for high-quality marijuana relative to low and medium quality remains in columns (4) and (6). Given that two of our main instruments, distance to Mexico City and electricity price, are state-level averages, we are unable to perform IV estimation with state dummies.

The Sargan–Hansen statistic in each IV specification fails to reject the null that our instruments, distance to Mexico City, the natural log of electricity price, and medical marijuana, are valid. These values are 2.46, 2.28, 1.95, and 1.83, for columns (4), (5), (6), and (7), respectively. The Cragg–Donald Wald F statistics for all of the IV specifications suggest that the instruments are strong.[9] Insofar as the medical marijuana instrument is theoretically weak for uniquely identifying supply, columns (5) and (7) report estimates using only distance to Mexico City and the log of electricity price as instruments. The price elasticity decreases in each of these as compared to columns (4) and (6), respectively. The lower of these point estimates, however, −0.59, is still consistent with the extant literature reviewed above. The coefficients on income and high quality also lose their significance.

4.3 Variation in price

While the main purpose of this paper was to estimate the price elasticity of demand, given the wide geographic coverage of these data and the information on quality, this section briefly explores variation in the price of marijuana by quality, region, and medical marijuana status. Particular attention is paid to any quality premium that may exist and how medical marijuana laws, which may increase both the demand and supply of marijuana, impact price. Finally, regional differences are also explored.

Table 4 provides descriptive statistics and tests for significant differences in price by quality, medical marijuana, and census region. Here again observations are weighted by the inverse of the probability that each state is in our sample and all statistical test are conducted using cluster robust standard errors. In terms of quality, high-quality

[9] The Cragg–Donald Wald F statistics are 108.46, 75.21, 115.26, and 68.26 for columns (4), (5), (6), and (7), respectively. These values indicate that none of the IV specifications suffer from weak instruments.

marijuana sells for an average price of $13.77 per gram, approximately $4.50 above medium quality and $8 above low quality, suggesting a substantial quality premium. The null that there is no difference in price by quality is firmly rejected. Table 6 in the Appendix provides elasticity estimates distinguished by quality. As in Table 3, the IV model yields slightly more elastic price elasticity estimates than OLS, but by and large the price elasticity is similar across quality, particularly in our IV specification.[10]

States with medical marijuana laws have different prices per gram than states without these laws, yet the difference depends on quality. For high-quality marijuana, medical marijuana states pay a lower price per gram, whereas for medium- and low-quality marijuana, the price is higher, albeit not statistically different. While speculative without further analysis, this suggests that any increases in demand for marijuana may have been offset by greater increases in supply for high-quality marijuana, a plausible outcome if marijuana provided for medical purposes is of higher quality. In contrast, the increased demand for medium- and lower-quality marijuana is not offset by corresponding increases in supply, resulting in higher prices and consistent with Pacula et al. (2010).

There is also evidence of significant price variation by region, with notably lower prices for high-quality marijuana in the Mountain ($11.85 per gram) and Pacific ($10.13 per gram) states relative to others, where prices range from $14.02 per gram in the East North Central states to $15.18 in the West South Central. Differences in prices between regions are statistically significant, even when excluding the Mountain and Pacific states.[11]

The prices differences for high-quality marijuana by region also exist for medium- and low-quality marijuana, but the most and least expensive regions change. For medium-quality marijuana, the East South Central states are the least expensive with the Mid-Atlantic and New England States being the most expensive. For low-quality marijuana the West South Central states are the least expensive, whereas the Pacific states, which have the lowest price for high quality, are the most expensive for low quality. Indeed, the spread between high- and low-quality prices is much lower in the Pacific region than other regions. Because quality is subjective, this may reflect the fact that in states where high-quality marijuana is relatively abundant (such as those on the Pacific coast) individuals may classify marijuana as low quality that which may be considered medium or high quality elsewhere.[12] While our data do not allow a determination of the reason for the lower spread in prices in the Pacific region, it is an area in need of additional research.

These results suggest that there is noticeable price variation by region. Reasons for this variation are not clear, and it is not the intention of this paper to fully explain

[10] Though nearly all of the results in Table 6 are consistent with our other findings, the high-quality estimates from the IV estimation with just two instruments lose significance. We suspect this is due to weighting and clustering on the state while also using only state-specific instruments.

[11] States corresponding with each Census region are as follows: New England (ME, NH, VT, MA, CT, RI); Mid-Atlantic (NY, PA, NJ); South Atlantic (WV, MD, DE, VA, NC, SC, GA, FL); East South Central (MS, AL, TN, KY); East North Central (WI, MI, IL, IN, OH); West North Central (ND,SD, NE, KS, MO, IA, MN); West South Central (OK, AR, LA, TX); Mountain (MT, ID, WY, UT, CO, AZ, NM, NV); and Pacific (WA, OR, CA).

[12] We are very grateful to an anonymous referee for bringing this issue and explanation to our attention.

Table 4 Variation in price by quality, medical marijuana law, and region[a]

	High	Medium	Low	*F* Test[b]
Quality				
Price per Gram	13.77	9.10	5.63	185.43***
	(4.74)	(5.96)	(5.47)	
	$n = 14{,}172$	$n = 7836$	$n = 1603$	
Medical Marijuana Law				
Price per Gram	Yes	No	***t-test***[c]	
(High quality)				
	12.56	14.48	3.39***	
	(4.53)	(4.77)		
	$n = 5112$	$n = 9060$		
Price per Gram	9.71	8.98	1.03	
(Medium quality)	(5.17)	(6.29)		
	$n = 2442$	$n = 5394$		
Price per Gram	6.76	5.94	0.99	
(Low quality)	(6.27)	(5.79)		
	$n = 332$	$n = 1271$		

Region									
New Eng.	Mid Atl	S. Atl	ES Cent	EN Cent	WS Cent	WN Cent	Mount	Pac	*F Test*[d]
Price per gram (high quality)									
14.19	14.90	15.11	14.50	14.02	15.18	14.45	11.85	10.13	16.10***
(4.06)	(4.73)	(4.99)	(5.90)	(4.62)	(4.92)	(4.10)	(3.91)	(3.85)	
$n = 1108$	$n = 2078$	$n = 2484$	$n = 434$	$n = 2057$	$n = 1070$	$n = 1071$	$n = 1162$	$n = 2708$	

Table 4 continued

Region New Eng.	Mid Atl	S. Atl	ES Cent	EN Cent	WS Cent	WN Cent	Mount	Pac	*F Test*[d]
Price per gram (Medium quality)									
10.72	10.70	8.34	6.65	8.70	7.08	10.42	10.09	9.17	15.06***
(5.75)	(6.27)	(6.03)	(4.86)	(6.90)	(6.16)	(6.38)	(5.08)	(3.98)	
$n = 650$	$n = 1294$	$n = 1479$	$n = 335$	$n = 1311$	$n = 541$	$n = 517$	$n = 437$	$n = 1272$	
Price per gram (Low quality)									
6.64	7.10	6.61	5.15	6.11	3.54	5.87	6.58	9.23	8.95***
(5.42)	(6.03)	(7.07)	(4.99)	(5.78)	(4.85)	(4.67)	(5.32)	(8.13)	
$n = 138$	$n = 216$	$n = 306$	$n = 80$	$n = 269$	$n = 261$	$n = 167$	$n = 81$	$n = 85$	

[a] Observations weighted by the inverse of the probability that each state is on our sample. Standard deviation in parentheses
[b] F test of null hypothesis that no difference in price by quality in a weighted and cluster robust regression of price per gram on dummies for each quality
[c] t test that coefficient on dummy for medical marijuana equals zero in weighted and cluster robust regression of price per gram on dummy for medical marijuana
[d] F test of null hypothesis that no difference in price by region in a weighted and cluster robust regression of price per gram on dummies for each region

regional variation. However, Caulkins (1995) found that marijuana prices increased as one went from the Midwest to the East, supporting his hypothesis that greater distance from domestic cultivation centers (the Midwest and Appalachia in his study) resulted in higher prices. A similar pattern arises here, with prices in New England and the Mid-Atlantic states generally higher than other regions. Similarly, the lower prices for medium- and low-quality marijuana in the West South Central states may reflect their proximity to Mexico. The lower prices for high-quality marijuana in the Pacific and Mountain states may reflect their proximity to Northern California. At this point, the correlation between cultivation centers and price is purely speculative and the ability to grow marijuana indoors likely weakens this correlation. Nevertheless, there appears to be significant regional price variation, the reason for which is worthy of further study.

5 Conclusions

Policies toward marijuana legalization are changing, evidenced most recently in Colorado, Washington, Alaska, and the District of Columbia, where voters approved marijuana legalization. Such policies are likely to significantly impact both the demand for and supply of marijuana. To understand the likely impact these changes will have on consumption, prices, and tax revenue, estimates of elasticity are needed.

This study used a unique data set that contained information on price, quantity, quality, and geographic location. All fifty states were represented in the sample, although for this analysis we excluded Alaska and Hawaii. This data set has many advantages including a large sample of marijuana transactions made by actual users rather than drug enforcement agents, information on quality, and a wide geographic representation. As a result, this is one of the few studies to provide conventional, full demand price elasticity estimates based on actual quantities purchased rather than the participation elasticity. These advantages, however, must be caveated with the recognition that the data are self-reported, resulting in quality differences that are subjective with no socioeconomic or frequency of purchase information available at the individual level. Moreover, the data are a cross section, thereby prohibiting any temporal analysis. Thus, like all marijuana data, this data set is clearly imperfect.

Unique as this study is in providing a current price elasticity of demand for marijuana using actual transitions data, the estimates of elasticity in this study are generally consistent with the few other extant studies. Our estimates place the price elasticity of demand between -0.67 and -0.79, with the former slightly above the elasticity estimates found in Australia by Clements and Daryal (2005), and Clements (2009), but consistent with Clements et al. (2010). Our range of elasticity estimates are slightly larger but not inconsistent with Clements (2008, 2009) finding that marijuana price elasticity estimates are minus one-half. These estimates are at the lower end of those reported by Nisbet and Vakil (1972) over forty years ago, though the age of their study alone is justification for new estimates. Thus, the demand for marijuana appears relatively insensitive to price changes. To the extent that, for example, medical marijuana or decriminalization laws increase supply and reduce price, consumption may be less responsive, *ceteris paribus*. Such policies may also significantly increase demand, necessitating further studies of the supply of marijuana similar to

Pacula et al. (2010). Indeed, this analysis found evidence of differential prices in marijuana between medical marijuana and non-medical marijuana states, with this differential depending on the quality reported, suggesting that changes in supply and demand may not be uniform across different types of marijuana. Of course, our elasticity estimates are based on demand for a product that was illegal in all states when the data were collected and our price estimate does not account for enforcement risk. Whether the price elasticity would remain the same should marijuana become legal remains unknown, although Colorado, Washington, Alaska, and the District of Columbia may provide interesting case studies. Until data from those jurisdictions or other legal settings become available, data sets such as the one used here, while imperfect, contribute to a better understanding of the market for marijuana.

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Appendix

see Tables 5 and 6.

Table 5 First-stage estimates (weighted and clustered IV) *Dependent variable: log of price per gram*

Variables	(1)	(2)	(3)	(4)
Ln (income)	0.4576***	0.4706***	0.4881***	0.5129***
	(0.1222)	(0.1423)	(0.1249)	(0.1443)
Ln (beer tax)	−0.0059	0.0026	−0.0071	0.0019
	(0.0225)	(0.0269)	(0.0219)	(0.0275)
Quality high	1.047***	1.037***	1.048***	1.036***
	(0.0571)	(0.0581)	(0.0570)	(0.0583)
Quality medium	0.4918***	0.4883***	0.4926***	0.4890***
	(0.0469)	(0.0466)	(0.0466)	(0.0462)
Decriminalized	−0.0635*	−0.0831**		
	(0.0329)	(0.0399)		
ln(electricity price)	0.0123	−0.0353	−0.0042	−0.0628
	(0.1205)	(0.1477)	(0.1210)	(0.1486)
Distance to Mexico City	0.0001***	0.00008**	0.0001***	0.00008**
	(0.00003)	(0.00004)	(0.00002)	(0.00004)
Medical Marijuana State	−0.1193***		−0.1289***	
	(0.0316)		(0.0339)	
Constant	−3.623***	−3.608***	−3.918***	−4.002***
	(1.117)	(1.263)	(1.148)	(1.288)
F of excluded instruments	13.77***	6.68***	13.47***	4.84**
N	23,611	23,611	23,611	23,611

Cluster robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 6 Weighted and cluster robust OLS and IV estimates of marijuana demand by quality *dependent variable: log of grams of marijuana purchased*

Variables	OLS (1)	OLS (2)	OLS[a] (3)	IV (4)	IV[b] (5)	IV (6)	IV[b] (7)
High quality ($n = 14, 172$)							
Ln (price)	−0.687***	−0.684***	−0.698***	−0.722***	−0.358	−0.719***	−0.437
	(0.0346)	(0.0336))	(0.0364)	(0.171)	(0.387)	(0.156)	(0.362)
Ln (income)	0.156	0.161		0.187*	0.0756	0.191*	0.101
	(0.103)	(0.0999)		(0.103)	(0.157)	(0.103)	(0.154)
Ln (beer tax)	0.0528*	0.0517*		0.0530**	0.0396	0.0526**	0.0424
	(0.0285)	(0.0283))		(0.0269)	(0.0340)	(0.0268)	(0.0326)
Decriminalized	−0.0422			−0.0413	−0.00162		
	(0.0359)			(0.0412)	(0.0517)		
Constant	2.376**	2.300**	4.006***	2.141**	2.359**	2.071**	2.294**
	(1.067)	(1.037)	(0.0918)	(1.010)	(1.004)	(0.976)	(1.015)
R^2	0.143	0.142	0.158				
Medium quality ($n = 7836$)							
Ln (price)	−0.699***	−0.699***	−0.689***	−0.718***	−0.721***	−0.722***	−0.723***
	(0.0253)	(0.0252)	(0.0126)	(0.0851)	(0.0855)	(0.0880)	(0.0883)
Ln (income)	0.138	0.144		0.140	0.144	0.150	0.152
	(0.105)	(0.106)		(0.119)	(0.119)	(0.120)	(0.120)
Ln (beer tax)	0.00758	0.144		0.00862	0.00842	0.00626	0.00615
	(0.0300)	(0.106)		(0.0312)	(0.0312)	(0.0310)	(0.0310)

Table 6 continued

Variables	OLS (1)	OLS (2)	OLS[a] (3)	IV (4)	IV[b] (5)	IV (6)	IV[b] (7)
High quality ($n = 14,172$)							
Decriminalized	−0.0416			−0.0482	−0.0482		
	(0.0327)			(0.0318)	(0.0318)		
Constant	2.193**	2.118*	3.554***	2.208*	2.179*	2.097*	2.082*
	(1.085)	(1.098)	(0.0466)	(1.169)	(1.166)	(1.183)	(1.182)
R^2	0.316	0.316	0.332				
Low quality ($n = 1603$)							
Ln (price)	−0.627***	−0.626***	−0.569***	−0.766***	−0.762***	−0.761***	−0.755***
	(0.0641)	(0.0646)	(0.0544)	(0.107)	(0.108)	(0.108)	(0.108)
Ln (income)	0.192	0.199		0.0437	0.0412	0.0224	0.0188
	(0.265)	(0.266)		(0.242)	(0.242)	(0.244)	(0.243)
Ln (beer tax)	0.0683*	0.0676*		0.0236	0.0239	0.0268	0.0272
	(0.0359)	(0.0366)		(0.0237)	(0.0237)	(0.0239)	(0.0239)
Decriminalized	−0.0382			0.0482	0.0482		
	(0.0738)			(0.0705)	(0.0704)		
Constant	1.626	1.547	3.620***	3.324	3.345	3.557	3.588
	(2.745)	(2.752)	(0.740)	(2.495)	(2.488)	(2.525)	(2.516)
R^2	0.252	0.252	0.323				

Cluster robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

[a] State dummies included

[b] IV estimates with only the natural log of electricity price and distance to Mexico City as instruments

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